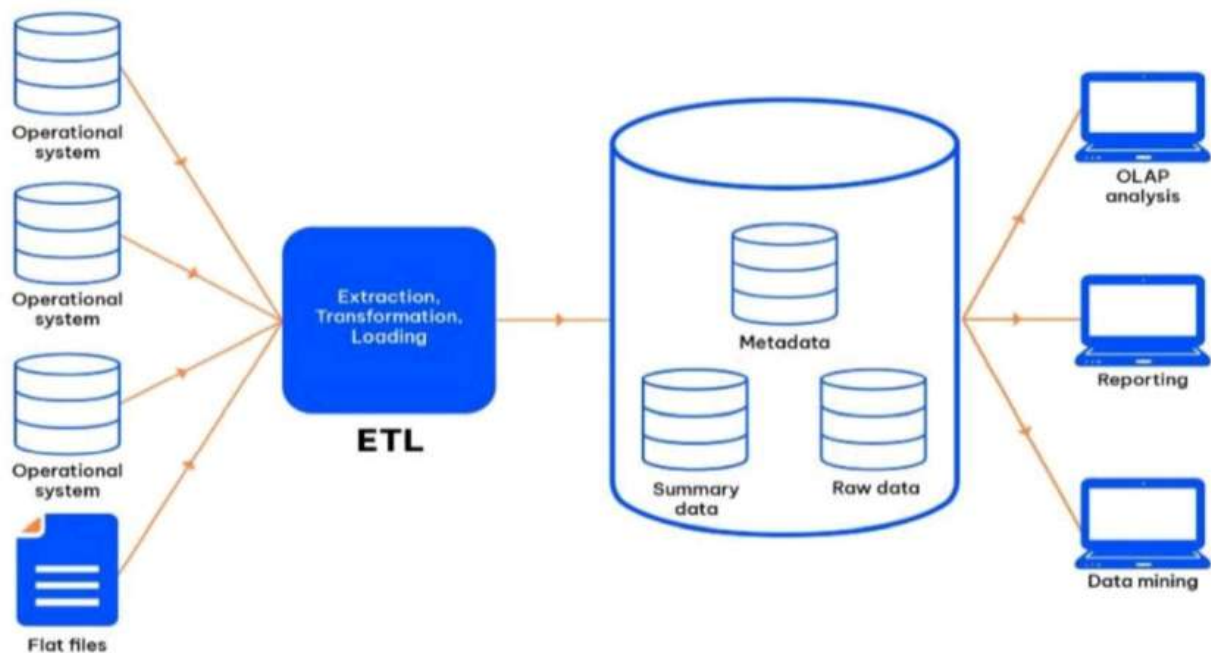


DATA WAREHOUSE — COMPLETE GUIDE FOR DATA ENGINEERS (2026)

Definition of Data Warehouse



1. What is a Data Warehouse? (Interview Definition)

A **Data Warehouse (DWH)** is a **centralized, read-optimized system** designed to store **historical, integrated data** from multiple sources for **analytics, reporting, and decision-making**.

It is **NOT** used for transactions

It is built for **analysis, trends, and KPIs**

One-line interview answer:

A data warehouse stores cleaned, historical business data optimized for analytical queries and reporting.

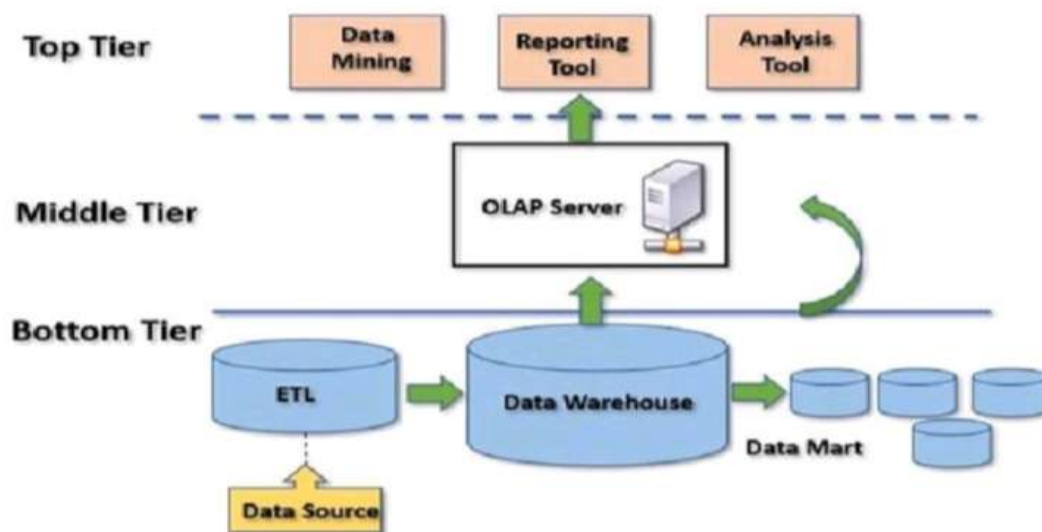
2. OLTP vs OLAP (MOST ASKED)

Feature	OLTP	OLAP
Purpose	Transactions	Analytics
Queries	Insert / Update	Complex SELECT
Data	Current	Historical
Tables	Normalized	Denormalized
Example	Order placement	Sales trend analysis

Interview Tip:

If you confuse OLTP & OLAP → interviewer knows fundamentals are weak.

3. Data Warehouse Architecture (3-Tier)



1. Source Layer

- Databases (MySQL, Oracle)
- APIs
- Files (CSV, JSON)
- Streaming sources

2. Staging Layer

- Raw data landing zone
- No transformations
- Used for validation & recovery

3. Presentation Layer

- Fact & Dimension tables
- Used by BI tools (Power BI, Tableau)

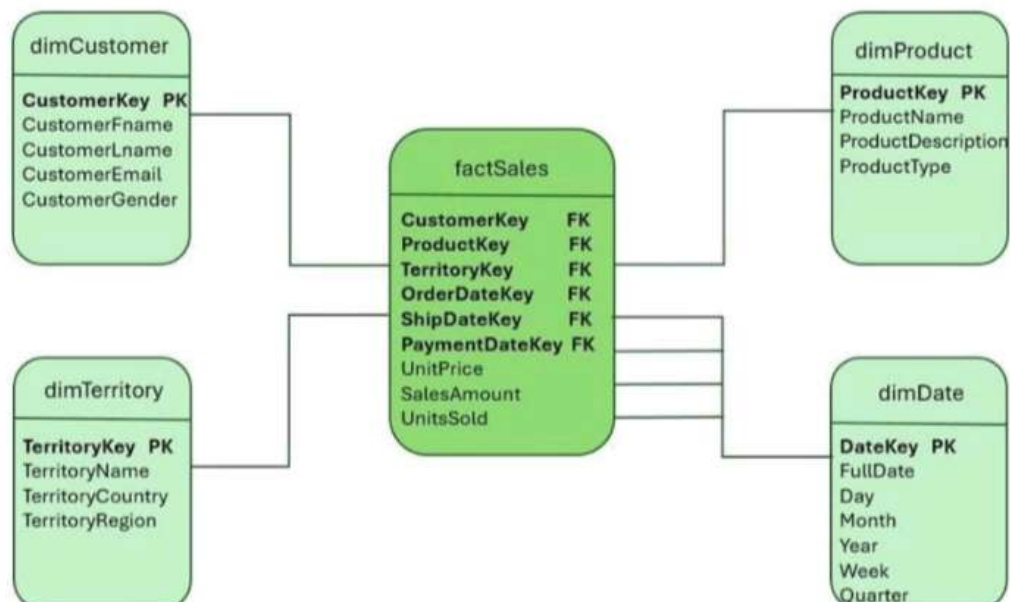
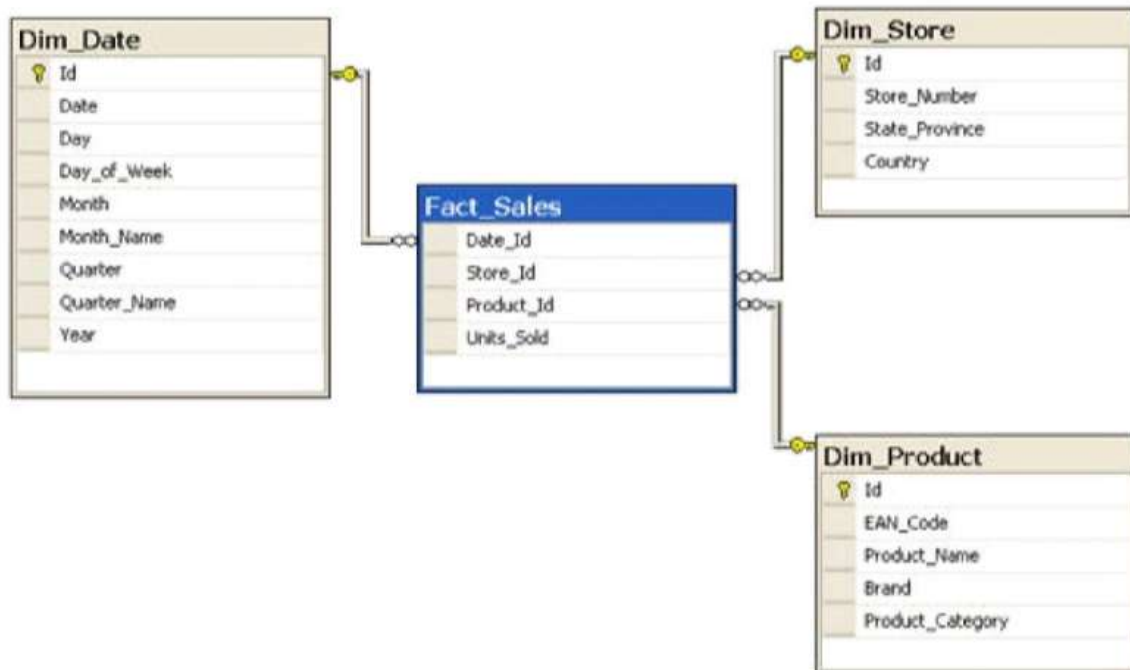
4. ETL vs ELT

ETL	ELT
Transform before load	Load first, transform later
Used in legacy DWH	Used in cloud DWH
Slower	Faster & scalable

Real Interview Example:

We used **ELT with Snowflake** because transformation was done using SQL after loading raw data.

5. Fact Tables (VERY IMPORTANT)



Types of Fact Tables

1. **Transactional Fact** – One row per transaction

2. **Snapshot Fact** – Daily / monthly snapshots
3. **Accumulating Snapshot** – Process lifecycle tracking

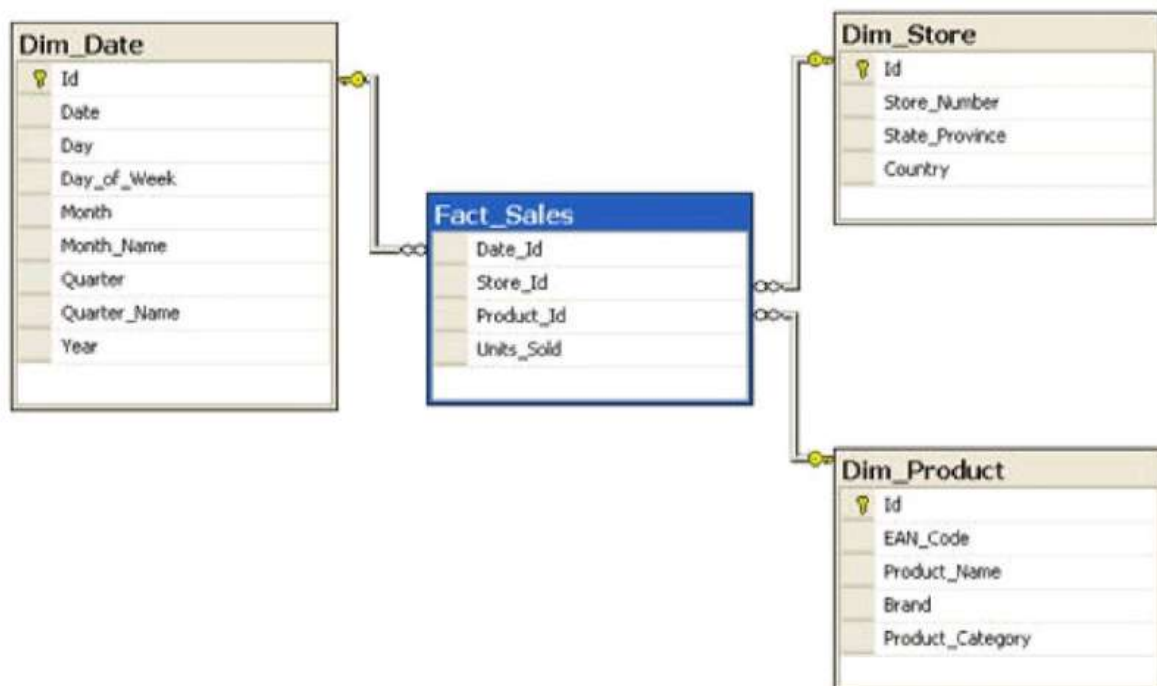
Fact Table Contains:

- Measures (Sales, Quantity)
- Foreign keys to dimensions

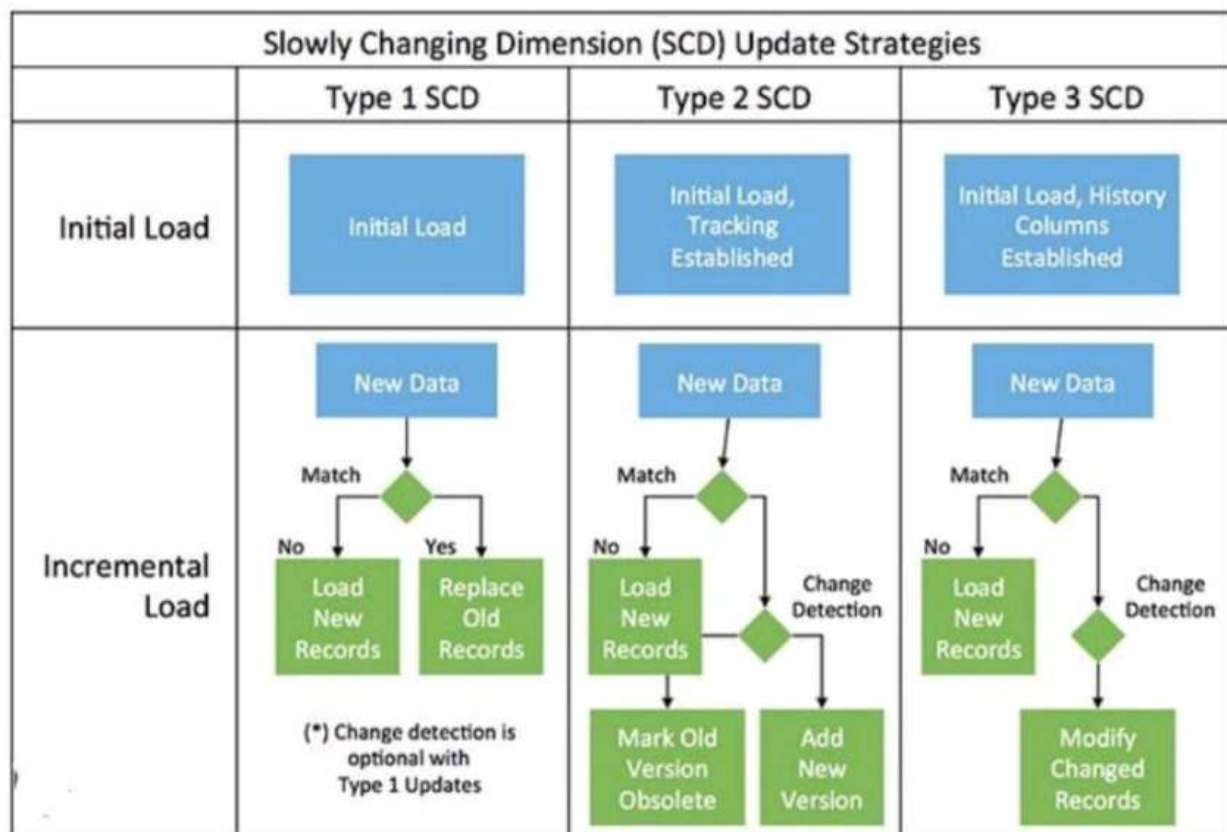
Interview Trick:

Facts = numeric + measurable

6. Dimension Tables



Tech Interview Titans



Common Dimensions

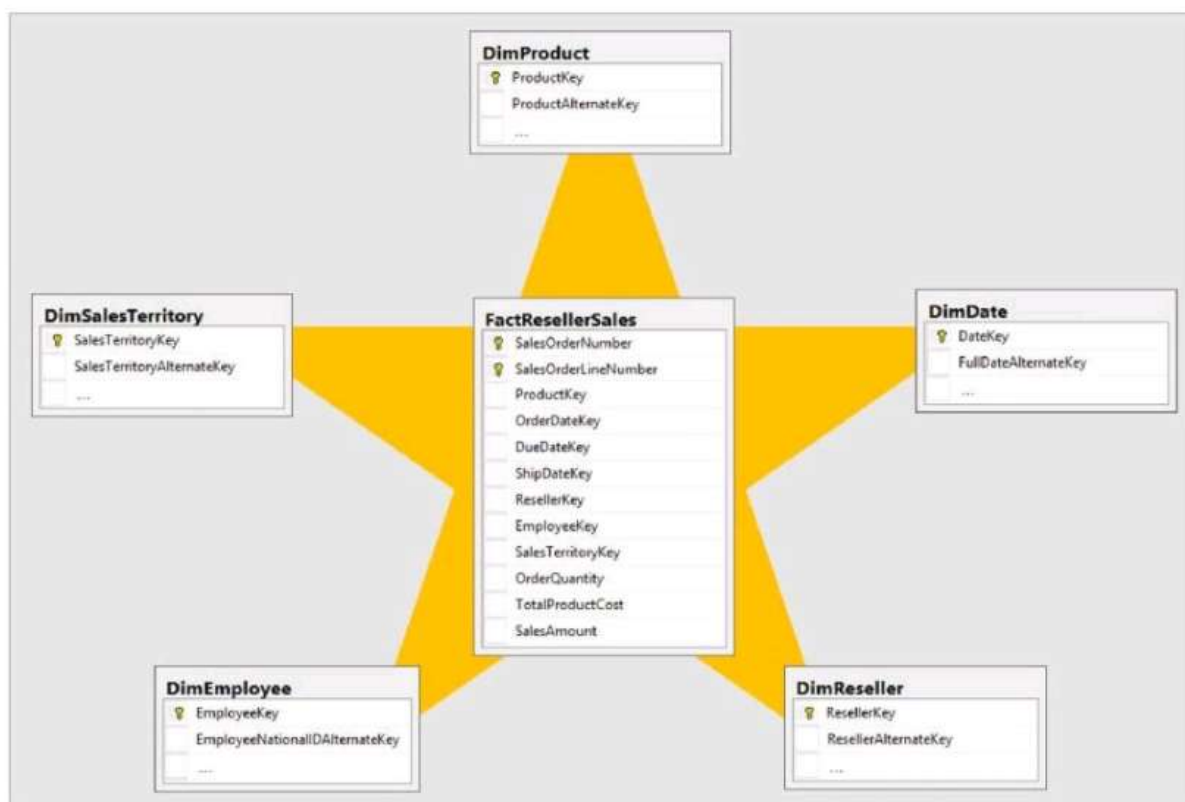
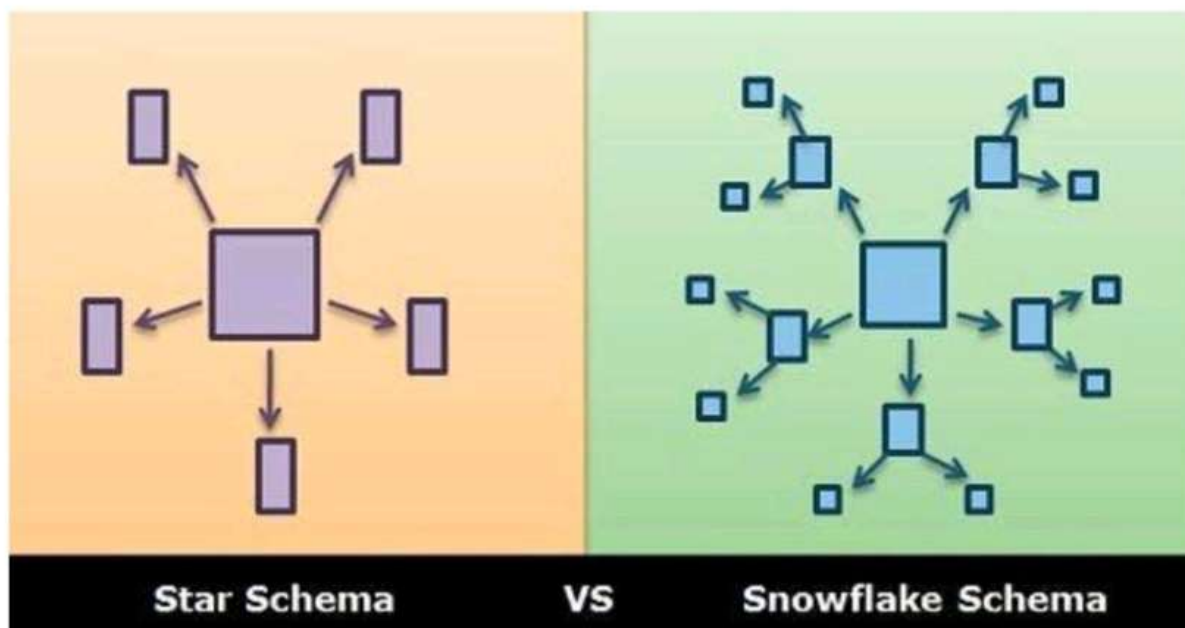
- Date
- Customer
- Product
- Location

Slowly Changing Dimensions (SCD)

Type	Meaning
SCD 0	No change
SCD 1	Overwrite
SCD 2	New row (history)
SCD 3	Limited history

MOST ASKED: SCD Type 1 vs Type 2

7. Star vs Snowflake Schema



Star Schema	Snowflake Schema
Simple	Complex
Faster queries	Slower

Denormalized	Normalized
Preferred	Rarely used

Interview Preference: **STAR SCHEMA**

8. Granularity (INTERVIEW)

Granularity = level of detail stored in fact table

Example:

- Order level → High granularity
- Daily sales → Low granularity

Once granularity is decided → **cannot be changed easily**

9. Surrogate Key vs Natural Key

Surrogate Key	Natural Key
System generated	Business key
Integer	String
Stable	Can change

Best Practice: Always use **Surrogate Keys**

10. Data Warehouse Load Types

1. **Full Load** – Load everything
2. **Incremental Load** – Load only changes
3. **CDC (Change Data Capture)** – Track inserts/updates/deletes

Incremental load is preferred in real projects.

11. Partitioning & Indexing

Partitioning

- Improves query performance

- Common on **date columns**

Indexing

- Speeds up search
- Avoid over-indexing on large fact tables

12. Data Quality Checks

- Null validation
- Duplicate detection
- Referential integrity
- Count reconciliation

Interview Reality:

Most production failures are **data quality issues**, not code.

TOP INTERVIEW QUESTIONS WITH ANSWERS (2026)

Q1. Why do we use a Data Warehouse?

Answer:

To analyze historical data across systems and support business decision-making.

Q2. Difference between Data Lake and Data Warehouse?

Answer:

Data Lake stores raw data; Data Warehouse stores structured, cleaned, analytical data.

Q3. What is SCD Type 2?

Answer:

It tracks historical changes by inserting a new row for every change.

Q4. What happens if fact table granularity is wrong?

Answer:

Reports become inaccurate and redesign is required.

Q5. Why use surrogate keys?

Answer:

Because natural keys can change and affect historical data consistency.

REAL INTERVIEW SCENARIO

Question:

Customer address changes — how do you track history?

Answer:

Use **SCD Type 2** with effective date and current flag.